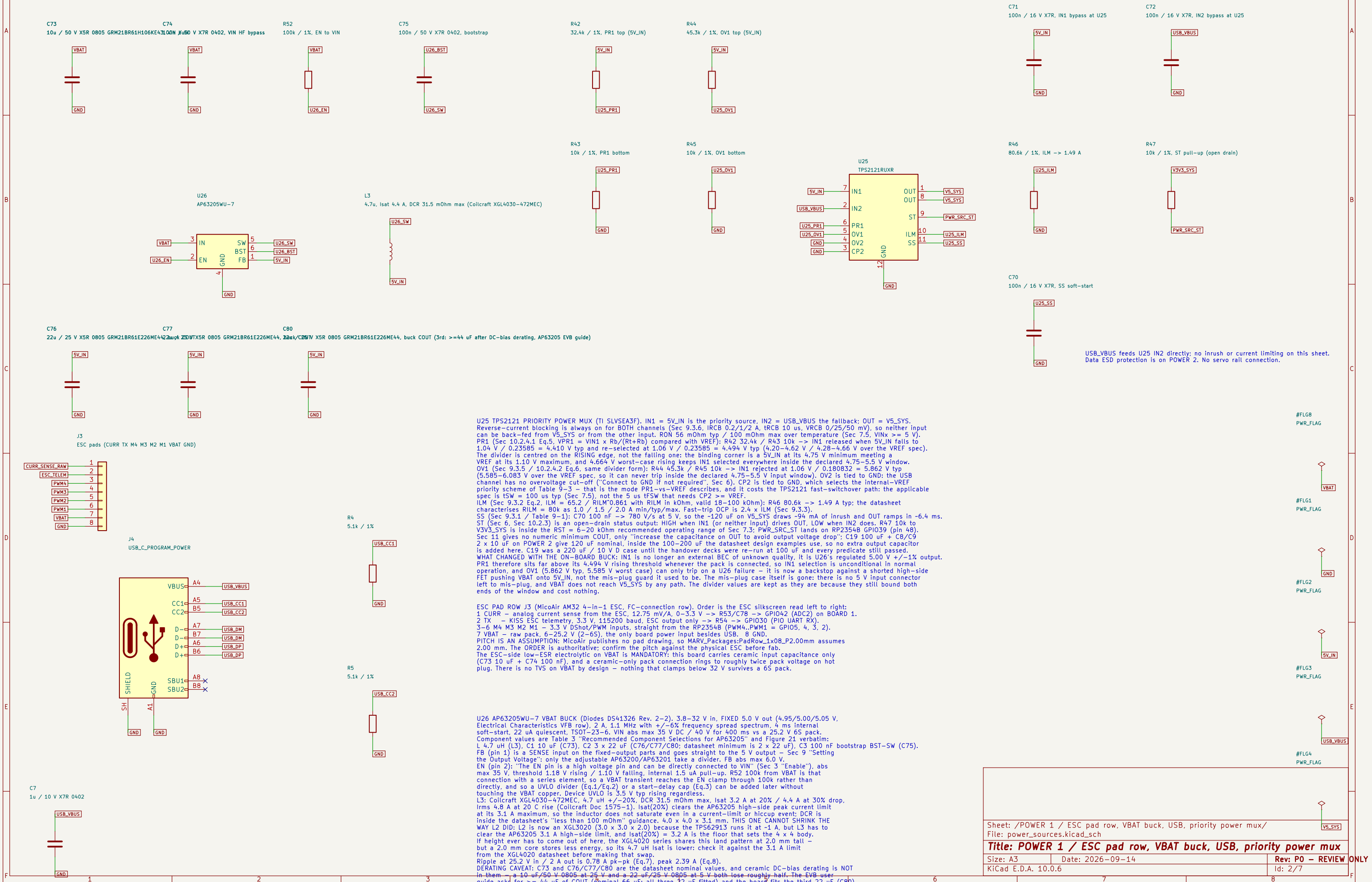


1	2	3	4	5	6	7	8	
A	Integrated draft: power chain (sheets 2–4) and board stage (sheets 5–8). See DESIGN_SPEC.md. Inputs: VBAT 6–25.2 V (2–6S) from the MicoAir AM32 ESC pad row J3, and USB VBUS. Rails: 5V_IN from the U26 AP63205 buck, V5_SYS from the U25 TPS2121 mux, V3V3_SYS from the TPS62913 buck, V3V3_ANA from the TPS7A20 LDO. Servo 5 V is 5V_IN, upstream of the mux. No PCB placement or routing has been performed.							A
B	POWER 1 / ESC pad row, VBAT buck, USB, priority power mux			BOARD 2 / IMU, barometer, high-g				B
	File: power_sources.kicad_sch			File: sensors.kicad_sch				
C	POWER 2 / 3.3 V system buck and analog LDO			BOARD 3 / log flash and microSD				C
	File: power_3v3.kicad_sch			File: storage.kicad_sch				
D	POWER 3 / IO array							D
	File: power_periph.kicad_sch							
E	BOARD 1 / RP2354B							E
	File: mcu.kicad_sch							
F								F
						Sheet: / File: MARV-V2.kicad_sch		
						Title: MARV integrated power – preliminary		
						Size: A3	Date: 2026-09-14	Rev: P0 – NOT FOR FAB
						KiCad E.D.A. 10.0.6	Id: 1/7	

BOARD INPUTS: VBAT 6–25.2 V (2–6S) from the ESC pad row J3, and USB VBUS ~5 V (J4). U26 (AP63205WU) makes 5V_IN from VBAT; 5V_IN is the priority input of the U25 power mux and also the servo 5 V bus.

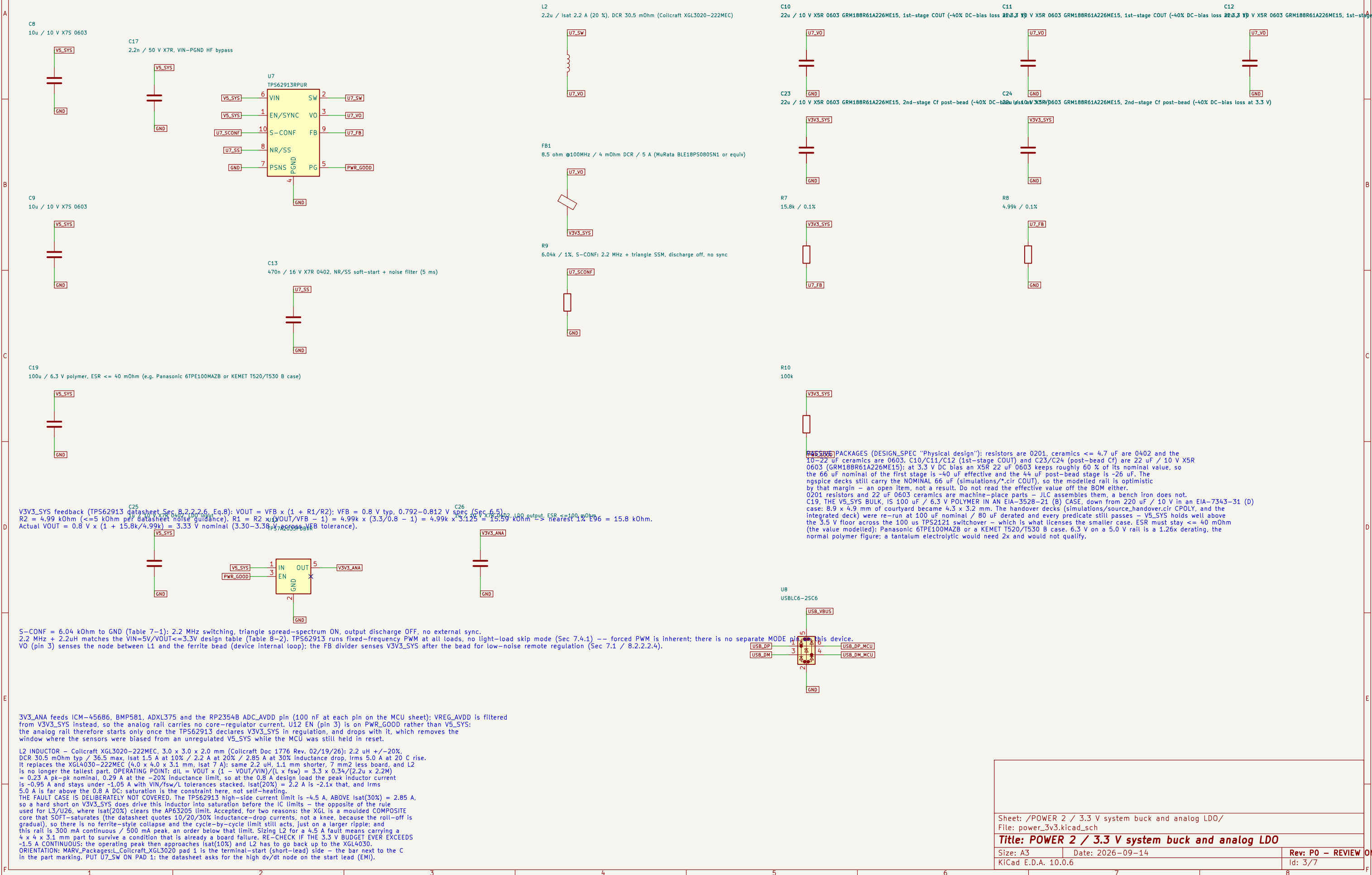


U25 TPS2121 PRIORITY POWER MUX (TI SLVSEA3F). IN1 = 5V_IN is the priority source, IN2 = USB_VBUS the fallback; OUT = V5_SYS. Reverse-current blocking is always on for BOTH channels (Sec 9.3.6, IRCB 0.2/1/2 A, tRCB 10 us, VRCB 0/25/50 mV), so neither input can be back-fed from V5_SYS or from the other input. RON 56 mOhm typ / 100 mOhm max over temperature (Sec 7.5, VINx >= 5 V). PR1 (Sec 10.2.4.1 Eq.5, VPR1 = VIN1 x Rb/(Rt+Rb) compared with VREF): R42 32.4k / R43 10k -> IN1 released when 5V_IN falls to 1.04 V / 0.23585 = 4.410 V typ and re-selected at 1.06 V / 0.23585 = 4.494 V typ (4.20–4.62 V / 4.28–4.66 V over the VREF spec). The divider is centred on the RISING edge, not the falling one: the binding corner is a 5V_IN at its 4.75 V minimum meeting a VREF at its 1.10 V maximum, and 4.664 V worst-case rising keeps IN1 selected everywhere inside the declared 4.75–5.5 V window. OV1 (Sec 9.3.5 / 10.2.4.2 Eq.6, same divider form): R44 45.3k / R45 10k -> IN1 rejected at 1.06 V / 0.180032 = 5.862 V typ (5.585–6.083 V over the VREF spec, so it can never trip inside the declared 4.75–5.5 V input window). OV2 is tied to GND: the USB channel has no overvoltage cut-off ("Connect to GND if not required"; Sec 6). CP2 is tied to GND, which selects the internal-VREF priority scheme of Table 9–3 – that is the mode PR1–vs–VREF describes, and it costs the TPS2121 fast-switchover path: the applicable spec is tSW = 100 us typ (Sec 7.5), not the 5 us tFSW that needs CP2 >= VREF. ILM (Sec 9.3.2 Eq.2, ILM = 65.2 / RILM*0.861 with RILM in kOhm, valid 18–100 kOhm): R46 80.6k -> 1.49 A typ: the datasheet characterises RILM = 80k as 1.0 / 1.5 / 2.0 A min/typ/max. Fast-trip OCP is 2.4 x ILM (Sec 9.3.3). SS (Sec 9.3.1 / Table 9–1): C70 100 nF -> 780 V/s at 5 V, so the ~120 uF on V5_SYS draws ~94 mA of inrush and OUT ramps in ~6.4 ms. ST (Sec 6, Sec 10.2.3) is an open-drain status output: HIGH when IN1 (or neither input) drives OUT, LOW when IN2 does. R47 10k to V3V3_SYS is inside the RST = 6–20 kOhm recommended operating range of Sec 7.3; PWR_SRC_ST lands on RP2354B GPIO39 (pin 48). Sec 11 gives no numeric minimum COUT; only "increase the capacitance on OUT to avoid output voltage drop"; C19 100 uF + C8/C9 2 x 10 uF on POWER 2 give 120 uF nominal, inside the 100–200 uF the datasheet design examples use, so no extra output capacitor is added here. C19 was a 220 uF / 10 V D case until the handover decks were re-run at 100 uF and every predicate still passed. WHAT CHANGED WITH THE ON-BOARD BUCK: IN1 is no longer an external BEC of unknown quality, it is U26's regulated 5.00 V +/-1% output. PR1 therefore sits far above its 4.494 V rising threshold whenever the pack is connected, so IN1 selection is unconditional in normal operation, and OV1 (5.862 V typ, 5.585 V worst case) can only trip on a U26 failure – it is now a backstop against a shorted high-side FET pushing VBAT onto 5V_IN, not the mis-plug guard it used to be. The mis-plug case itself is gone: there is no 5 V input connector left to mis-plug, and VBAT does not reach V5_SYS by any path. The divider values are kept as they are because they still bound both ends of the window and cost nothing.

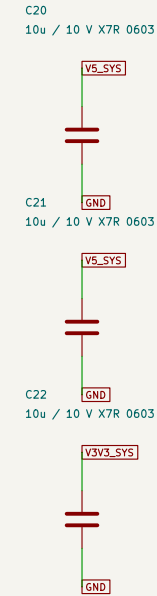
ESC PAD ROW J3 (MicoAir AM32 4–in–1 ESC, FC-connection row). Order is the ESC silkscreen read left to right: 1 CURR – analog current sense from the ESC, 12.75 mV/A, 0–3.3 V -> R53/C78 -> GPIO42 (ADC2) on BOARD 1. 2 TX – KISS ESC telemetry, 3.3 V, 115200 baud, ESC output only -> R54 -> GPIO30 (PIO UART RX). 3–6 M4 M3 M2 M1 – 3.3 V DShot/PWM inputs, straight from the RP2354B (PWM4, PWM1 = GPIO5, 4, 3, 2). 7 VBAT – raw pack, 6–25.2 V (2–6S), the only board power input besides USB. 8 GND. PITCH IS AN ASSUMPTION: MicoAir publishes no pad drawing, so MARV_Packages:PadRow_1x08_P2.00mm assumes 2.00 mm. The ORDER is authoritative; confirm the pitch against the physical ESC before fab. The ESC-side low-ESR electrolytic on VBAT is MANDATORY: this board carries ceramic input capacitance only (C73 10 uF + C74 100 nF), and a ceramic-only pack connection rings to roughly twice pack voltage on hot plug. There is no TVS on VBAT by design – nothing that clamps below 32 V survives a 6S pack.

U26 AP63205WU–7 VBAT BUCK (Diodes DS41326 Rev. 2–2), 3.8–32 V in, FIXED 5.0 V out (4.95/5.00/5.05 V, Electrical Characteristics VFB row), 2 A, 1.1 MHz with +/-6% frequency spread spectrum, 4 ms internal soft-start, 22 uA quiescent, ISOT–23–6, VIN abs max 35 V DC / 40 V for 400 ms vs a 25.2 V 6S pack. Component values are Table 3 "Recommended Component Selections for AP63205" and Figure 21 verbatim: L 4.7 uH (L3), C1 10 uF (C73), C2 3 x 22 uF (C76/C77/C80; datasheet minimum is 2 x 22 uF), C3 100 nF bootstrap BST–SW (C75). FB (pin 1) is a SENSE input on the fixed-output parts and goes straight to the 5 V output – Sec 9 "Setting the Output Voltage": only the adjustable AP63200/AP63201 take a divider. FB abs max 6.0 V. EN (pin 2): "The EN pin is a high voltage pin and can be directly connected to VIN" (Sec 3 "Enable"), abs max 35 V, threshold 1.18 V rising / 1.10 V falling, internal 1.5 uA pull-up, R52 100k from VBAT is that connection with a series Vbat, so a VBAT transient reaches the EN clamp through 100k rather than directly, and so a UVLO divider (Eq.1/Eq.2) or a start-delay cap (Eq.3) can be added later without touching the VBAT copper. Device UVLO is 3.5 V typ rising regardless. L3: Coilcraft XGL4030–472MEC, 4.7 uH +/-20%, DCR 31.5 mOhm max, Isat 3.2 A at 20% / 4.4 A at 30% drop, Irms 4.8 A at 20 C rise (Coilcraft Doc 1575–1). Isat(20%) clears the AP63205 high-side peak current limit at its 3.1 A maximum, so the inductor does not saturate even in a current-limit or hiccup event; DCR is inside the datasheet's "less than 100 mOhm" guidance. 4.0 x 4.0 x 3.1 mm. THIS ONE CANNOT SHRINK THE WAY L2 DID: L2 is now an XGL3020 (3.0 x 3.0 x 2.0) because the TPS62913 runs it at ~1 A, but L3 has to clear the AP63205 3.1 A high-side limit, and Isat(20%) = 3.2 A is the floor that sets the 4 x 4 body. If height ever has to come out of here, the XGL4020 series shares this land pattern at 2.0 mm tall – but a 2.0 mm core stores less energy, so its 4.7 uH Isat is lower: check it against the 3.1 A limit from the XGL4020 datasheet before making that swap. Ripple at 25.2 V in / 2 A out is 0.78 A pk–pk (Eq.7), peak 2.39 A (Eq.8). DERATING CAVEAT: C73 and C76/C77/C80 are the datasheet nominal values, and ceramic DC-bias derating is NOT in them – a 10 uF/50 V 0805 at 25 V and a 22 uF/25 V 0805 at 5 V both lose roughly half. The EVB user guide asks for >= 44 uF of COUT (nominal 66 uF; all three 22 uF fitted) and the board fits the third 22 uF (C80). PACKAGES: C73 is a 10 uF/50 V X5R 0805 (GRM21BR61H106KE43) and C76/C77/C80 are 22 uF/25 V X5R 0805 (GRM21BR61E226ME44) – down from 1206 for the 50 mm single-sided envelope; C74/C75 are 50 V X7R 0402. Same capacitance, same voltage rating, smaller body: the DC-bias loss above is the 0805 figure and is not made worse by the package change at these ratings, but none of these parts is bench-reworkable.

AVIONICS ONLY: 300 mA continuous / 500 mA short peak, provisional. NO SERVO POWER.

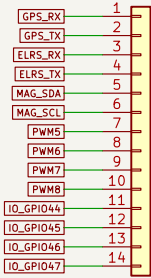


IO ARRAY – ONE 3-COLUMN x 14-ROW THROUGH-HOLE BLOCK ON THE RIGHT EDGE, 2.54 mm, madflitght FC3v2 style. It replaces the old 2x16 left-edge array AND the J12/J13/J14 servo block: every signal that leaves this board except the ESC row (J3) and the SWD pads (J10) now leaves on this one block. COLUMN RULE, from the board edge inward: GND (J8) | POWER (J7) | SIGNAL (J6). The SIGNAL column is the INNERMOST one, next to the MCU, so the 14 signal traces are the short ones; the two rails run down the outside where a plane stitch is cheap. A 3-wire module lead (signal, power, ground) plugs straight across ONE row. THREE Conn_01x14 SYMBOLS, not one 3x14: no 3-row generic connector symbol ships with this KiCad install, and three 1x14 rows with a body-trimmed courtyard abut exactly on the 2.54 mm grid (see tools/setup_pcb.py). DS-009 NAMING IS KEPT: a port label is named for the PERIPHERAL pin it belongs to, so GPS_RX / ELRS_RX are module inputs driven by the RP2354B UART TX pins (rows 1 and 3, silk T0 / T1) and GPS_TX / ELRS_TX are module outputs that land on the MCU UART RX pins (rows 2 and 4, silk R0 / R1).

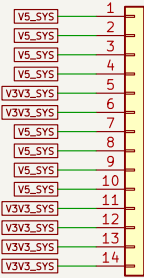


C20 / C21: V5_SYS local bulk at the array (rows 1–4 GPS/ELRS and rows 7–10 servo, the 5 V supply rows).
C22: V3V3_SYS local bulk at the array (rows 5–6 and 11–14).

J6
IO array signal column (innermost)



J7
IO array power column (5V / 3V3 per row)



J8
IO array GND column (board edge)



IO ARRAY ROW MAP (row n = J6 pin n signal | J7 pin n power | J8 pin n GND)

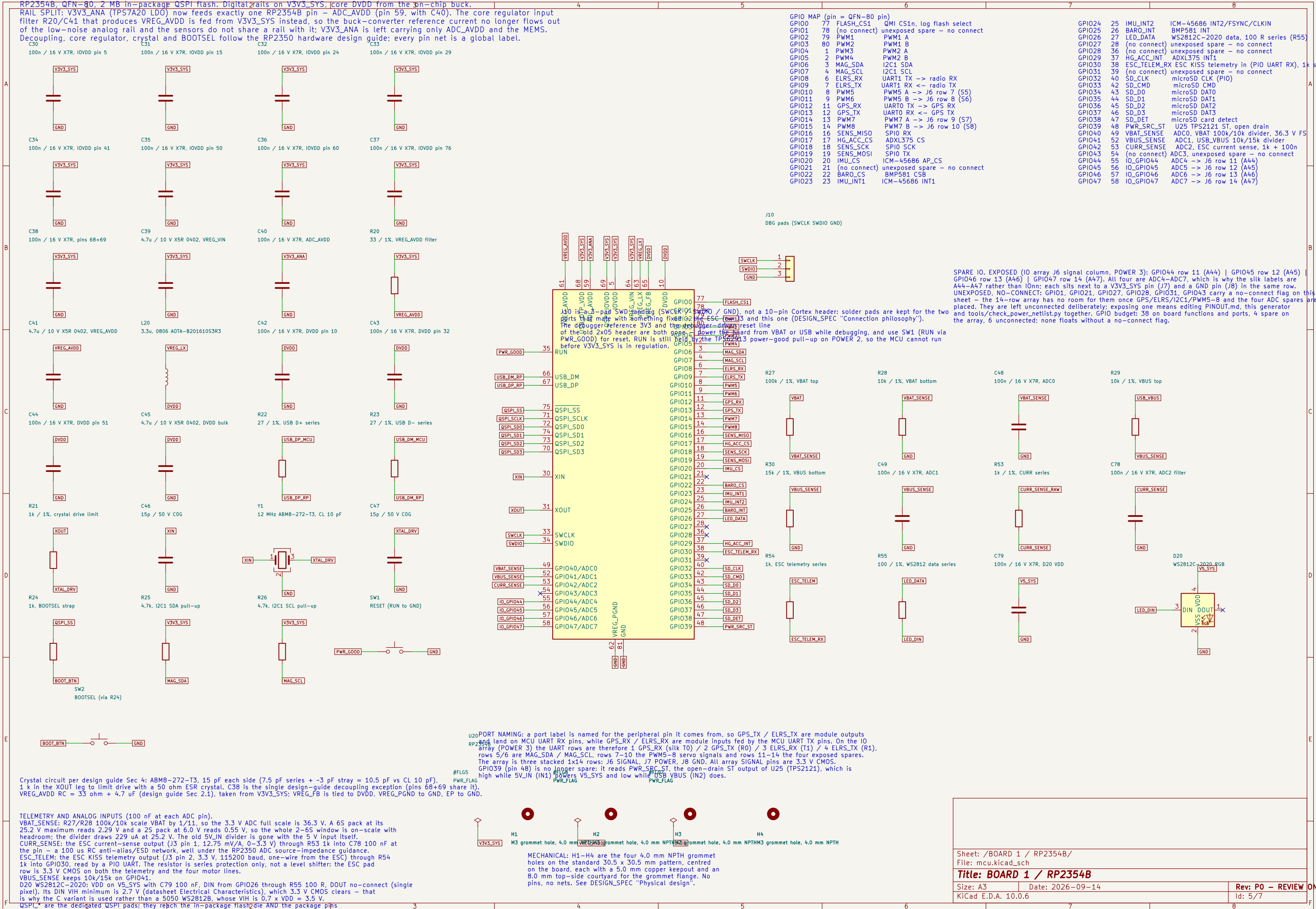
1	T0	GPS_RX	V5_SYS	GND	UART0 TX -> GPS RX (GPIO12)
2	R0	GPS_TX	V5_SYS	GND	UART0 RX <- GPS TX (GPIO13)
3	T1	ELRS_RX	V5_SYS	GND	UART1 TX -> ELRS RX (GPIO8)
4	R1	ELRS_TX	V5_SYS	GND	UART1 RX <- ELRS TX (GPIO9)
5	SDA	MAG_SDA	V3V3_SYS	GND	I2C1 SDA (GPIO6), 4.7k pull-up on the MCU sheet
6	SCL	MAG_SCL	V3V3_SYS	GND	I2C1 SCL (GPIO7), 4.7k pull-up on the MCU sheet
7	S5	PWM5	V5_SYS	GND	PWM5 A (GPIO10), servo 5
8	S6	PWM6	V5_SYS	GND	PWM5 B (GPIO11), servo 6
9	S7	PWM7	V5_SYS	GND	PWM7 A (GPIO14), servo 7
10	S8	PWM8	V5_SYS	GND	PWM7 B (GPIO15), servo 8
11	A44	IO_GPI044	V3V3_SYS	GND	GPIO44 = ADC4, spare
12	A45	IO_GPI045	V3V3_SYS	GND	GPIO45 = ADC5, spare
13	A46	IO_GPI046	V3V3_SYS	GND	GPIO46 = ADC6, spare
14	A47	IO_GPI047	V3V3_SYS	GND	GPIO47 = ADC7, spare

PWM MAP
PWM5 = GPIO10 (row 7, S5)
PWM6 = GPIO11 (row 8, S6)
PWM7 = GPIO14 (row 9, S7)
PWM8 = GPIO15 (row 10, S8)

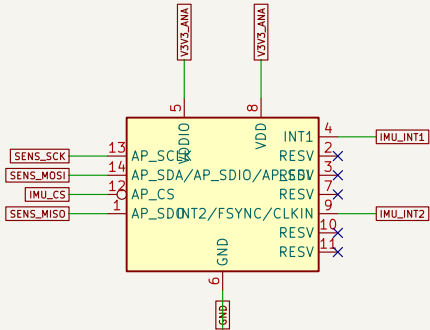
ON THE ESC PAD ROW (J3, POWER 1)
PWM1 = GPIO2
PWM2 = GPIO3
PWM3 = GPIO4
PWM4 = GPIO5

RAILS ON THE ARRAY: rows 1–4 (GPS, ELRS) and rows 7–10 (servos S5–S8) take V5_SYS; rows 5–6 (magnetometer) and rows 11–14 (the four exposed spares) take V3V3_SYS. Every row has its own GND pin in the J8 column, so a signal return is never more than 2.54 mm away and no row can be mis-plugged into a neighbour's ground. SERVO 5 V NOW COMES FROM V5_SYS, NOT 5V_IN. The old J13 servo row tapped 5V_IN directly at the U26 buck output, ahead of the U25 priority mux, so servo current was outside the mux current limit. With one power column the array cannot carry two different 5 V nets under one "5V" silk label, so all eight 5 V rows are V5_SYS: servo current now crosses U25 and counts against its ILM limit (1.49 A, R46) and against the U26 buck. Budget servo current against BOTH – see DESIGN_SPEC "Rails and load budget". CURRENT: rows 1–4 and 7–10 count against the U26 buck through the mux; rows 5–6 and 11–14 count in the V3V3_SYS budget. No fuse and no per-row current limit: the array is a system-integration connector, not a protected port.

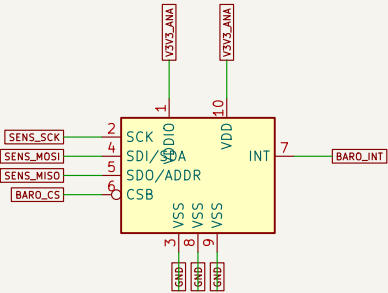
RP2354B, QFN-80, 2 MB in-package QSPI flash. Digital rails on V3V3_SYS, core DVDD from the on-chip buck.
RAIL SPLIT: V3V3_ANA (TPS7A20 LDO) now feeds exactly one RP2354B pin – ADC_AVDD (pin 59, with C40). The core regulator input filter R20/C41 that produces VREG_AVDD is fed from V3V3_SYS instead, so the buck-converter reference current no longer flows out of the low-noise analog rail and the sensors do not share a rail with it; V3V3_ANA is left carrying only ADC_AVDD and the MEMS. Decoupling, core regulator, crystal and BOOTSEL follow the RP2350 hardware design guide; every pin net is a global label.



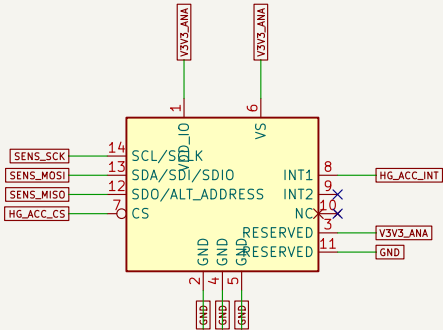
All three sensors run 4-wire SPI on the shared SENS_SCK / SENS_MOSI / SENS_MISO bus with one chip select each, and are powered from V3V3_ANA (TP57A20 LDO). Every supply pin on this sheet gets 100 nF + 1 uF; BMP581 and ADXL375 because their own datasheets call for the pair, the ICM-45686 as a deliberate addition – its datasheet BOM (DS-000577 Rev 1.0 Table 11) lists 0.1 uF only.



U21
ICM-45686 6-axis IMU



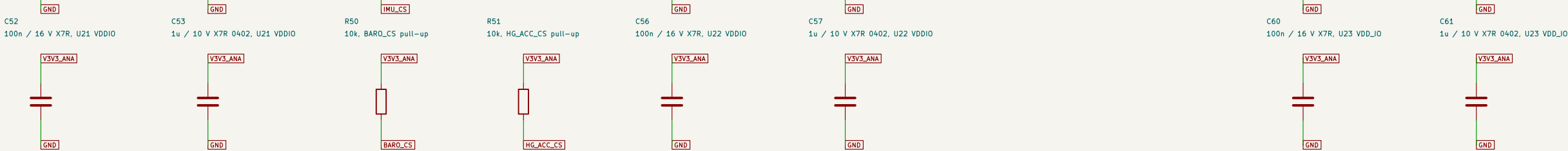
U22
BMP581 barometer



U23
ADXL375 high-g accelerometer



TEST POINTS TP1-TP10 (TestPoint:TestPoint_Pad_1.0x1.6mm, 1 x 1 mm bare F.Cu pads, 50 model by design – tools/audit footprints by --no-3d-ok covers TestPoint, along with the pad rows and mounting holes). TP1 SENS_SCK | TP2 SENS_MOSI | TP3 SENS_MISO | TP4 IMU_CS | TP5 BARO_CS | TP6 HG_ACC_CS | TP7 IMU_INT1 | TP8 IMU_INT2 | TP9 BARO_INT | TP10 HG_ACC_INT. Placement: Inside the sensor cluster, within 6 mm of U21/U22/U23 (tools/setup_pcb.py checks it). They add ten stubs to the SPI bus, which is why they are 1 mm pads next to the parts rather than a header somewhere else: at 10 MHz SCK a short stub is a capacitive load, a long one is a transmission line. No series resistors and no ground pad of their own – probe ground comes off a mounting hole or the J6 array.



U21 ICM-45686 (TDK InvenSense DS-000577 Rev 1.0, Table 10 "Signal Descriptions" / Figure 4 pin-out, Single Interface SPI mode, Figure 10): AP_CS (pin 12) = IMU_CS, AP_SCLK (13) = SENS_SCK, AP_SDA/AP_SDIO/AP_SDI (14) = SENS_MOSI, AP_SDO/AP_ADO (1) = SENS_MISO, INT1 (4) = IMU_INT1, INT2/FSYNC/CLKIN (9) = IMU_INT2 – typed bidirectional because FSNC/CLKIN are host-driven inputs while INT2 is a chip output; only the INT2 function is used here. VDD (8) and VDDIO (5) are both on V3V3_ANA; GND (6) is the only ground pin. Pins 2, 3, 7, 10 and 11 (RESV) are all left No Connect; Table 10 allows No Connect / VDDIO / GND for every RESV pin, and the Single-Interface SPI typical application schematic (Figure 10) shows all five left unterminated (internal pull enabled by default, per the per-pin notes in Table 10); they are only needed for the AUX1 or I2C-master dual-interface modes, neither of which this board uses. C50 (VDD) and C52 (VDDIO) are the 100 nF X7R of Table 11 "Bill of Materials" (C1/C2 in the datasheet); C51 (VDD) and C53 (VDDIO) add 1 uF on top of them. The datasheet BOM lists 0.1 uF only; this board keeps 100 nF + 1 uF per supply pin for consistency with U22/U23 and common flight-controller practice.

U22 BMP581: CSB = BARO_CS, SDI = SENS_MOSI, SDO = SENS_MISO, SCK = SENS_SCK, INT = BARO_INT, VDDIO and VDD on V3V3_ANA, all three VSS pins to GND.
U23 ADXL375: CS = HG_ACC_CS, SDA/SDI = SENS_MOSI, SDO = SENS_MISO, SCL/SCLK = SENS_SCK, INT1 = HG_ACC_INT. Per the pin function table, pin 3 (RESERVED) goes to VS and pin 11 (RESERVED) to GND; pin 10 (NC) and the unused INT2 are left open. VS and VDD I/O are both on V3V3_ANA.

R48, R50, R51: 10k pull-ups to V3V3_ANA on the three SPI chip selects (IMU_CS, BARO_CS, HG_ACC_CS). R49 (the former second IMU chip-select pull-up) is removed along with that net – the ICM-45686 has a single chip select. Same rationale as R35 on FLASH_CS1 (BOARD 3): every sensor is held deselected before the RP2354B drives the pin, so a GPIO left floating through reset, BOOTSEL or a debugger halt cannot open a transaction on the shared SENS_SCK / SENS_MOSI / SENS_MISO bus. They bias to V3V3_ANA, the same rail as every VDDIO pin on this sheet, so there is no pull-up current path into an unpowered I/O supply and no CS pin is pulled above its own VDDIO during rail sequencing.

Sheet: /BOARD 2 / IMU, barometer, high-g/
File: sensors.kicad_sch

Title: BOARD 2 / IMU, barometer, high-g

Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-14

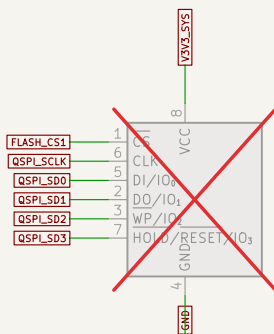
Rev: P0 – REVIEW ONLY

Id: 6/7

An OPTIONAL QSPI expansion socket on the shared QSPI bus with GPIO0/QM1 CS1n as its chip select, and a latched microSD wired for 4-bit PIO (also SPI-compatible). Both on V3V3_SYS with local bulk, per the rail split in DESIGN_SPEC.

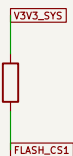
U24, C62 AND C63 ARE DNP – NOT FITTED BY DEFAULT AND NOT IN THE BOM. The land is on the BACK of the board (B.Cu) as a user expansion; firmware and logs use the RP2354B in-package 2 MB flash and the J11 microSD.

U24 IS A SOCKET, NOT ONE PART: the 150-mil SOIC-8 land takes either a W25Q64JVSSIQ NOR flash or an APS6404L-3SQR-SN 8 MB QSPI PSRAM - see the U24 note below. R35 stays POPULATED.



U24

DNP: W25Q64JVSSIQ, 64 Mbit QSPI NOR (optional back-side expansion)



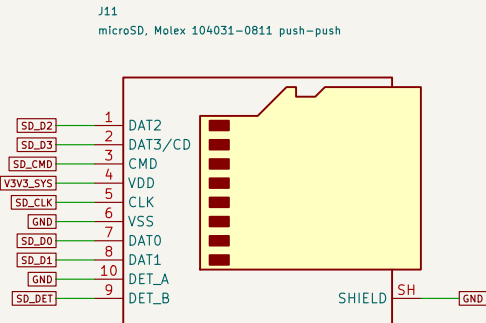
C62

100n / 16 V X7R, U24 VCC (fitted; socket is optional) / 10 V X7R 0402, U24 VCC (fitted; socket is optional)

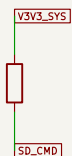


C63

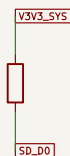
Small / 10 V X7R 0402, U24 VCC (fitted; socket is optional)



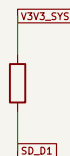
R36
10k, SD_CMD pull-up



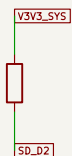
R37
10k, SD_D0 pull-up



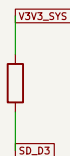
R38
10k, SD_D1 pull-up



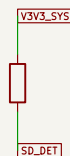
R39
10k, SD_D2 pull-up



R40
10k, SD_D3 pull-up



R41
10k, SD_DET pull-up



C64
100n / 16 V X7R, at the socket



C65
10u / 10 V X7R 0603, at the socket



C66
47u / 6.3 V X5R 0805, at the socket



U24 shares the dedicated QSPI pads with the RP2354B in-package flash die (datasheet Sec 14.3): CLK = QSPI_SCLK, DI/IO0 = QSPI_SDO, DO/IO1 = QSPI_SDI, WP/IO2 = QSPI_SD2, HOLD/IO3 = QSPI_SD3. Only the chip select differs - FLASH_CS1 comes from GPIO0 (QMI CS1n), pulled up to V3V3_SYS by R35 so the part is deselected before the MCU drives it. QSPI_SS (the internal die) keeps its own strap on the MCU sheet.

U24 SOCKET – OPTIONAL PSRAM OR SECOND FLASH. The land is Package_SO:SOIC-8_3.9x4.9mm_P1.27mm (150 mil), which fits BOTH the default W25Q64JVSSIQ (Winbond, 64 Mbit = 8 MB NOR flash, SOIC-8 150 mil) and the APS6404L-35QR-SN (AP Memory, 64 Mbit = 8 MB QSPI pseudo-SRAM, SOP-8 150 mil). The two are pin-identical on this bus: 1 CE#/CS#, 2 S0/SIO1, 3 SIO2 (WP# on the flash), 4 VSS/GND, 5 S1/SIO0, 6 SCLK, 7 SIO3 (HOLD# on the flash), 8 VCC – so no board change is needed to swap them. The RP2350 QM1 supports a PSRAM on CS1 with its own timing/chip-select registers (RP2350 datasheet Sec 12.14 QM1, M1_TIMING/M1_RFMT/M1_WFMT), which is what makes the alternative real rather than mechanical. Fit the W25Q64 flash for loq space or the PSRAM for

The alternative RNA-DNA interaction mechanism: Fit the W23Q49 class for toy-space or the "PSRAM" ($\text{X1A} + \Delta\text{G}\Delta\text{S}(\text{CAG})\text{GAA} - \Delta\text{H}\Delta\text{F}(\text{GATG})\text{GTTC}$) without Dk pull-up, so the input reads low with a card inserted. The DNPII stacks GAKTSKIGLZ2Q Gnt-Gnt-CGS early (B) sites for the (GmCmT)(G)GNSRPa feather, S(b)metamts RCG are the top board, thick parts-a-gate. Not Gd(b)dn dNtDk b(e)at eThe s(e)st and n(d)s i(n) the s(e)st and d(a)p(s)e c(e)ss. BAC3, Gnd(h)gh for the M(C)gdp d(e)l W203QS2Y3yads 70–75, so the SQSI stubs are the board thickness plus a via; the front stays a single-sided assembly because nothing DNP is reflowed. A B.SilKS legend beside the land says what fits.

CONSEQUENCE FOR THE BUDGET: the 25 mA QSPI-flash burst is NOT in the V3V3_SYS load budget any more (DESIGN_SPEC "Rails and load budget"); add it back if the socket is populated.

SYMBOL SUBSTITUTION: KiCad 10 ships no W25Q64JVSS symbol (Memory_Flash has W25Q16JVSS, W25Q32JVSS, W25Q128JV* only). U24 therefore¹ uses Memory_Flash:W25Q32JVSS, whose pinout is² identical across the whole W25Q JV family; the value field carries the real MPN. See LIBRARIES.md.³

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File: storage.kicad_sch

Title: BOARD 3 / log flash and microSD

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